sEMG Muscle Activation Report

All Participants · Gesture_1 · April 17, 2026

This document contains individual muscle-activation reports for each participant. Each report summarises per-sensor activation, spatial patterns, motor-unit recruitment indicators, visualisation, and methodology. Comparisons across participants should be interpreted with caution: without %MVC normalisation, absolute amplitude values are not directly comparable between subjects.

sEMG Muscle Activation Report

Participant P1 · Gesture_1 · April 17, 2026

Overall Activation	High
Temporal Profile	Ramp-down
Consistency	Consistent
Dominant Sensors	S4
Weakest Sensors	S7
Assessment Confidence	0.95

Clinical Summary: Sustained contraction shows high flexor activation, particularly in S4. The ramp-down profile in S4 suggests potential fatigue or loss of force maintenance over the 20-minute session. Co-contraction ratio is below the 0.5 threshold, indicating minimal antagonistic interference.

Key Findings:
- Ramp-down profile in S4 and S7
- High flexor-extensor imbalance

Session Overview

Session Duration	1207.6 s (20.1 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.008686 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.004786	0.170	105.6	80.2	22	Moderate	Sustained
S2	0.006875	0.067	106.5	85.0	126	High	Sustained
S3	0.015222	0.071	98.1	74.8	2128	Very High	Sustained
S4	0.023874	0.134	91.1	79.9	4176	Very High	Ramp-down
S5	0.005268	0.129	94.8	65.7	14	Moderate	Sustained
S6	0.005849	0.149	87.2	54.6	107	Moderate	Sustained
S7	0.003284	0.144	120.5	92.5	12	Moderate	Ramp-down
S8	0.004332	0.219	120.7	94.5	66	Moderate	Sustained

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S4: Primary driver, significant amplitude decay

S8: Borderline consistency

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S7	Lowest activity
Dominant-to-Weakest Ratio	7.27×	Activation disparity
Concentration Index	0.201	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.998	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.379	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.369	Agonist/antagonist balance
Flexor/Extensor Balance	2.73	>1 = flexor-dominant

Overall Temporal Profile	Ramp-down	Session-wide activation trend

Co-contraction: Not detected — Reciprocal activation pattern typical of efficient single-direction movement.

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Strong flexor dominance (S1-S4) compared to extensor group (S5-S8) with a flexor-extensor balance ratio of 2.73.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S4, S3

Evidence: High WAMP values in S4 (4176.3) and S3 (2127.5) indicate significant motor unit recruitment in the flexor group.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P2 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S2
Weakest Sensors	S6
Assessment Confidence	0.95

Clinical Summary: The session shows a stable, sustained contraction with high flexor-dominant activity. S2 is the primary driver of the gesture. Low inter-channel correlation suggests minimal crosstalk. The ramp-down in S4 and ramp-up in S8 indicate a shift in neuromuscular strategy over the 20-minute duration.

Key Findings:
- Very high flexor activation
- Flexor-extensor imbalance
- S4 ramp-down and S8 ramp-up suggest potential shift in recruitment strategy

Session Overview

Session Duration	1202.5 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.010762 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.011414	0.042	86.1	73.6	328	Very High	Sustained
S2	0.022872	0.068	91.7	80.8	4365	Very High	Sustained
S3	0.017858	0.106	98.3	82.9	2738	Very High	Sustained
S4	0.017739	0.144	115.6	105.5	3328	Very High	Ramp-down
S5	0.004217	0.188	132.3	111.2	57	Moderate	Sustained
S6	0.002850	0.077	139.4	114.2	8	Low	Sustained
S7	0.002975	0.053	139.1	115.8	17	Low	Sustained
S8	0.006174	0.077	113.5	94.7	95	Moderate	Ramp-up

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S2: Highest RMS amplitude

S4: Ramp-down profile observed

S6: Lowest RMS amplitude

S8: Ramp-up profile observed

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S2	Primary contributor
Weakest Sensor	S6	Lowest activity
Dominant-to-Weakest Ratio	8.03×	Activation disparity
Concentration Index	0.185	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.998	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.240	Crosstalk check (>0.8 = concern)

Co-Contraction Ratio	0.232	Agonist/antagonist balance
Flexor/Extensor Balance	4.32	>1 = flexor-dominant
Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Not detected — Reciprocal activation pattern typical of efficient single-direction movement.

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Strong flexor dominance (S1-S4) compared to extensor group (S5-S8) with a flexor-extensor balance ratio of 4.319.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S2, S4

Evidence: High WAMP values in S2 (4364.8) and S4 (3327.6) indicate significant motor unit recruitment during the sustained contraction.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

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sEMG Muscle Activation Report

Participant P3 · Gesture_1 · April 17, 2026

Overall Activation	High
Temporal Profile	Ramp-up
Consistency	Consistent
Dominant Sensors	S4
Weakest Sensors	S6
Assessment Confidence	0.95

Clinical Summary: Sustained isometric contraction with clear flexor dominance. S3 and S4 demonstrate high recruitment intensity. S8 shows a late-phase amplitude reduction, potentially indicating localized fatigue.

Key Findings:
- High flexor-extensor imbalance (2.23x)
- S3/S4 very high activation levels
- S8 amplitude decline in late phase

Session Overview

Session Duration	1202.5 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.006665 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.003516	0.142	109.8	79.3	5	Moderate	Sustained
S2	0.004810	0.388	109.2	86.5	74	High	Sustained
S3	0.012931	0.181	143.3	122.8	2944	Very High	Ramp-up
S4	0.015383	0.139	102.9	85.0	2206	Very High	Ramp-up
S5	0.005326	0.239	126.7	103.5	274	High	Sustained
S6	0.002719	0.127	111.3	74.4	5	Moderate	Sustained
S7	0.003254	0.170	90.8	51.0	7	Moderate	Ramp-up
S8	0.005384	0.246	79.7	56.3	40	High	Ramp-down

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S2: Moderate consistency

S3: High recruitment

S4: Primary driver

S5: Moderate consistency

S8: Late phase amplitude drop

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S6	Lowest activity
Dominant-to-Weakest Ratio	5.66×	Activation disparity
Concentration Index	0.185	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.994	Spatial consistency (1 = stable)

Inter-Channel Correlation	0.095	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.461	Agonist/antagonist balance
Flexor/Extensor Balance	2.23	>1 = flexor-dominant
Overall Temporal Profile	Ramp-up	Session-wide activation trend

Co-contraction: Not detected — Reciprocal activation pattern typical of efficient single-direction movement.

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows significantly higher activation (Very High) compared to extensor group (S5-S8), consistent with palmar flexion dominance.

Motor-Unit Recruitment

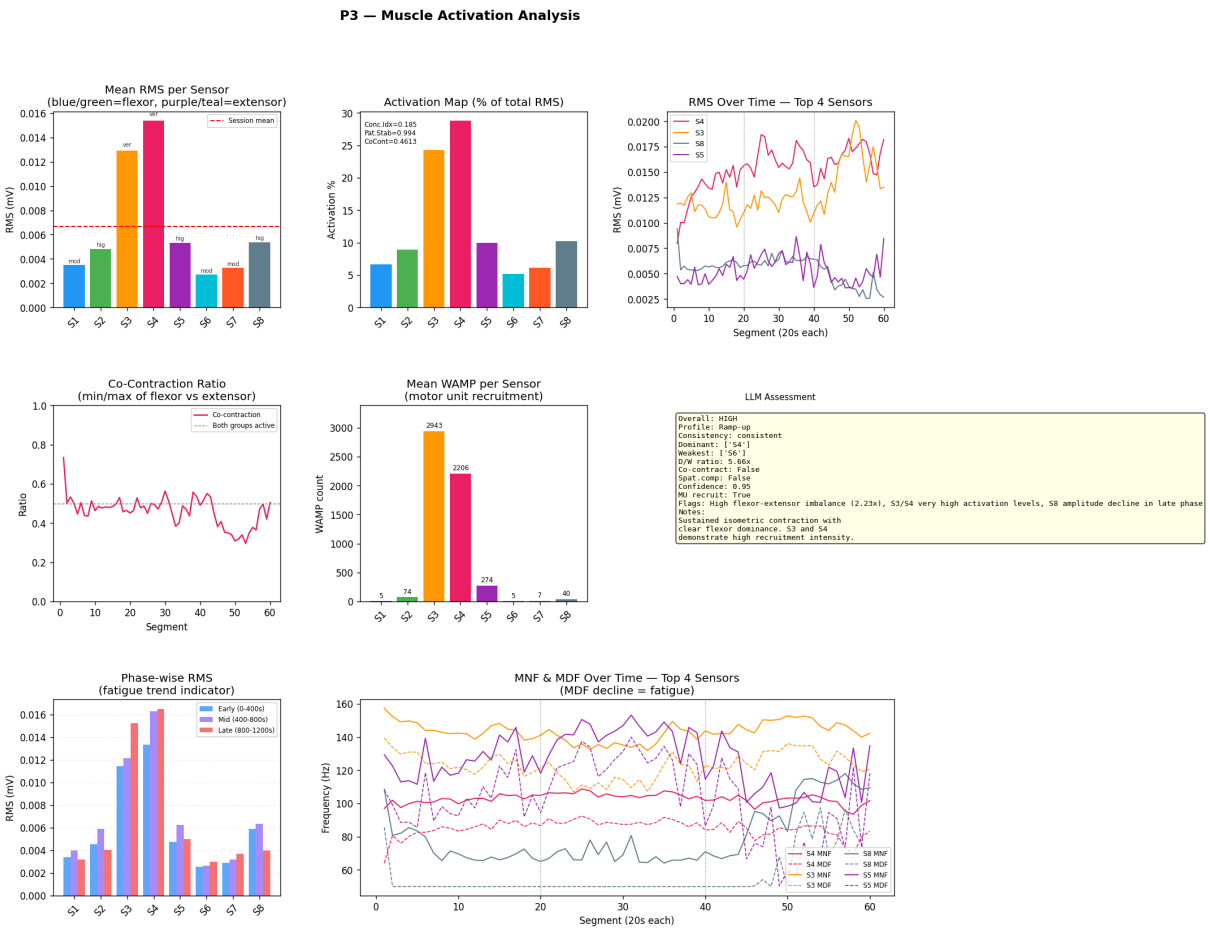
Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S3, S4

Evidence: High WAMP values (S3: 2943.8, S4: 2206.2) indicate significant motor unit recruitment in flexor group.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.



Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P4 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Ramp-up
Consistency	Moderate
Dominant Sensors	S4, S8
Weakest Sensors	S5, S6
Assessment Confidence	0.9

Clinical Summary: The session exhibits a clear ramp-up activation profile with high co-contraction, suggesting a power-grip or high-effort task. Motor unit recruitment indicators are elevated in S8 and S4. The high inter-channel stability suggests a consistent, albeit fatiguing, neuromuscular strategy.

Key Findings:
- High co-contraction ratio (>0.5)
- Significant ramp-up profile in S4 and S8
- High variability in S3

Session Overview

Session Duration	1202.4 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.019717 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.018533	0.374	106.8	91.8	1538	High	Sustained
S2	0.017613	0.275	134.2	125.4	3035	High	Sustained
S3	0.023059	0.663	57.8	50.0	1662	Very High	Intermittent
S4	0.031965	0.286	56.9	50.3	2651	Very High	Ramp-up
S5	0.011992	0.186	119.3	99.8	1421	Moderate	Sustained
S6	0.013168	0.381	108.8	89.4	1854	Moderate	Ramp-up
S7	0.015352	0.217	93.4	75.5	1571	High	Sustained
S8	0.026058	0.260	96.4	86.6	4876	Very High	Ramp-up

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S3: High variability in activation

S4: Primary dominant sensor

S8: High WAMP indicates significant recruitment

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S5	Lowest activity
Dominant-to-Weakest Ratio	2.67×	Activation disparity
Concentration Index	0.149	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.993	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.212	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.747	Agonist/antagonist balance

Flexor/Extensor Balance	1.36	>1 = flexor-dominant
Overall Temporal Profile	Ramp-up	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor-dominant activation (1.362 ratio). S4 (flexor) and S8 (extensor) show significant ramp-up profiles, suggesting increasing effort over the session duration.

Motor-Unit Recruitment

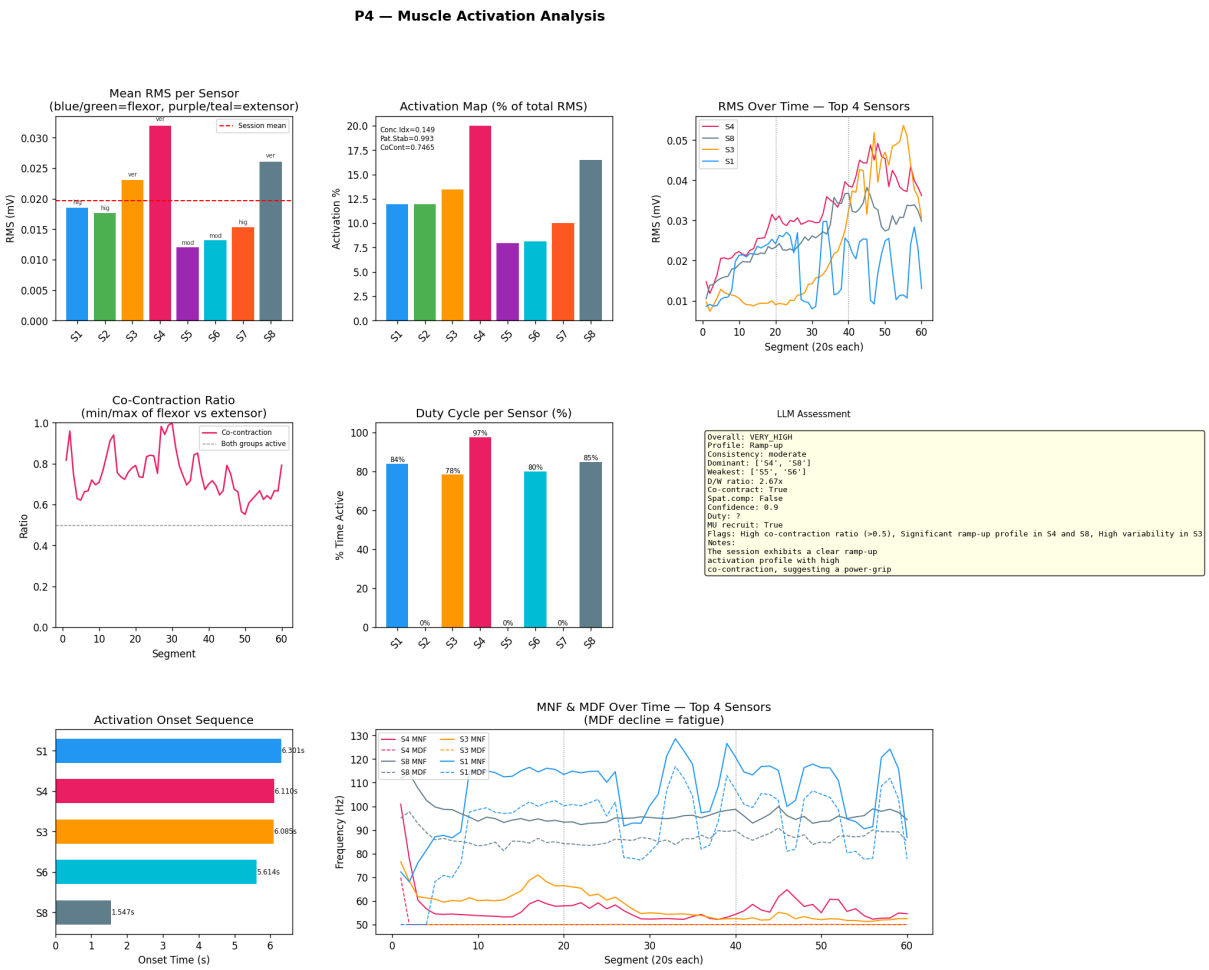
Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S8, S2, S4

Evidence: High WAMP values observed: S8 (4875.6), S2 (3035.1), and S4 (2651.3) suggest increased motor unit recruitment consistent with high-effort or fatigue-related compensation.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.



Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P5 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S3
Weakest Sensors	S7
Assessment Confidence	0.9

Clinical Summary: The session demonstrates a stable, sustained contraction with clear flexor dominance (S3). High WAMP values across multiple sensors suggest significant motor unit recruitment. Co-contraction ratio (0.5793) exceeds the 0.5 threshold, indicating simultaneous agonist/antagonist activity consistent with a power grip.

Key Findings:
- High co-contraction ratio
- Very high activation in S3
- Flexor-dominant activation pattern

Session Overview

Session Duration	1201.9 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.015157 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.004608	0.110	111.2	85.4	20	Moderate	Sustained
S2	0.008019	0.169	110.7	91.1	294	Moderate	Sustained
S3	0.045862	0.151	58.4	50.0	7979	Very High	Sustained
S4	0.019018	0.057	87.1	60.0	2612	Very High	Sustained
S5	0.018818	0.217	135.0	120.1	4333	Very High	Sustained
S6	0.015807	0.387	119.8	101.3	3238	Very High	Sustained
S7	0.004545	0.212	118.6	95.4	60	Low	Sustained
S8	0.004578	0.284	120.5	98.0	79	Moderate	Sustained

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S3: Primary driver of gesture

S4: Highly consistent

S6: Moderate consistency

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S3	Primary contributor
Weakest Sensor	S7	Lowest activity
Dominant-to-Weakest Ratio	10.09×	Activation disparity
Concentration Index	0.225	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.996	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.013	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.579	Agonist/antagonist balance

Flexor/Extensor Balance	1.89	>1 = flexor-dominant
Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows higher overall excitation compared to extensor group (S5-S8), with a flexor-extensor balance ratio of 1.8871.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S3, S5, S6

Evidence: High WAMP values observed in S3 (7978.8), S5 (4332.8), and S6 (3238.5) indicate significant motor unit recruitment during sustained contraction.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P6 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S4
Weakest Sensors	S2
Assessment Confidence	0.95

Clinical Summary: The session demonstrates a stable, high-intensity sustained contraction with significant co-contraction between flexor and extensor groups. S4 is the primary driver of the gesture. S8 shows a notable decrease in amplitude over time, while S6 shows a compensatory ramp-up, suggesting potential localized fatigue or shift in motor control strategy.

Key Findings:
- High co-contraction ratio
- S8 ramp-down profile
- S6 ramp-up profile

Session Overview

Session Duration	1202.4 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.019991 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.012043	0.220	120.1	102.1	1448	Moderate	Sustained
S2	0.008446	0.096	108.0	88.0	330	Moderate	Sustained
S3	0.018561	0.157	99.8	87.7	3026	High	Sustained
S4	0.040835	0.197	111.5	99.2	10311	Very High	Sustained
S5	0.027778	0.162	137.6	126.9	4985	Very High	Sustained
S6	0.011569	0.246	97.2	68.8	975	Moderate	Ramp-up
S7	0.014804	0.129	99.7	64.0	1836	High	Sustained
S8	0.025892	0.346	66.7	51.5	2195	Very High	Ramp-down

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S4: Highest WAMP/RMS

S6: Increasing activation trend

S8: Decreasing activation trend

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S2	Lowest activity
Dominant-to-Weakest Ratio	4.83×	Activation disparity
Concentration Index	0.162	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.996	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.222	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.855	Agonist/antagonist balance

Flexor/Extensor Balance	1.02	>1 = flexor-dominant
Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor (S1-S4) and Extensor (S5-S8) groups show balanced activation (ratio 1.0187), consistent with co-contraction during power grip.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S4, S5

Evidence: S4 shows highest WAMP (10310.7) and S5 shows high WAMP (4984.9), indicating significant motor unit recruitment during sustained contraction.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P7 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S3
Weakest Sensors	S8
Assessment Confidence	0.9

Clinical Summary: Sustained isometric contraction with clear flexor dominance. High co-contraction ratio (0.5376) suggests significant antagonist activity, typical for power grip. S3 is the primary active site with high recruitment indicators.

Key Findings:
- High flexor-extensor co-contraction ratio
- S1 intermittent activation instability
- Flexor dominance

Session Overview

Session Duration	1203.2 s (20.1 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.007758 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.009401	1.138	75.9	57.5	55	Very High	Intermittent
S2	0.010444	0.091	75.9	63.4	152	Very High	Sustained
S3	0.011533	0.100	80.4	67.5	424	Very High	Ramp-up
S4	0.009230	0.111	73.5	51.3	50	Very High	Ramp-up
S5	0.006764	0.079	107.9	82.7	271	High	Sustained
S6	0.004434	0.173	116.0	94.1	50	Moderate	Sustained
S7	0.006158	0.157	97.3	80.0	107	High	Sustained
S8	0.004104	0.267	97.8	70.5	21	Moderate	Sustained

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S1: High CV indicates instability in intermittent activation

S3: Highest RMS; primary driver

S8: Highest CV in extensor group

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S3	Primary contributor
Weakest Sensor	S8	Lowest activity
Dominant-to-Weakest Ratio	2.81×	Activation disparity
Concentration Index	0.146	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.991	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.088	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.538	Agonist/antagonist balance

Flexor/Extensor Balance	1.90	>1 = flexor-dominant
Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows higher activation levels (Very High) compared to extensor group (S5-S8, High/Moderate). Flexor-extensor balance of 1.90 confirms flexor dominance.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S3

Evidence: S3 exhibits highest WAMP (423.6), indicating significant motor unit recruitment during the sustained contraction.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P8 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Ramp-up
Consistency	Consistent
Dominant Sensors	S3
Weakest Sensors	S8
Assessment Confidence	0.95

Clinical Summary: The session demonstrates a stable, sustained contraction with a clear flexor-dominant pattern. High WAMP values in the flexor group (S2-S4) suggest robust motor unit recruitment. The low inter-channel correlation (0.2949) indicates minimal crosstalk risk.

Key Findings:
- High flexor-extensor imbalance
- Very high activation levels in S2-S5

Session Overview

Session Duration	1204.3 s (20.1 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.011663 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.004912	0.097	136.4	119.3	71	Moderate	Ramp-up
S2	0.014204	0.115	113.0	99.3	2169	Very High	Ramp-up
S3	0.031357	0.101	116.8	107.2	6325	Very High	Sustained
S4	0.014199	0.145	129.1	117.4	2814	Very High	Ramp-up
S5	0.013489	0.206	118.8	106.8	2279	Very High	Ramp-up
S6	0.006775	0.102	115.4	97.6	406	Moderate	Sustained
S7	0.004346	0.115	123.5	103.8	68	Moderate	Sustained
S8	0.004024	0.070	136.8	118.0	30	Moderate	Sustained

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S3: Primary driver of activation

S5: Higher variability observed

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S3	Primary contributor
Weakest Sensor	S8	Lowest activity
Dominant-to-Weakest Ratio	7.79×	Activation disparity
Concentration Index	0.194	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.998	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.295	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.444	Agonist/antagonist balance
Flexor/Extensor Balance	2.28	>1 = flexor-dominant

Overall Temporal Profile	Ramp-up	Session-wide activation trend

Co-contraction: Not detected — Reciprocal activation pattern typical of efficient single-direction movement.

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows significantly higher activation (Very High) compared to extensor group (S5-S8), with a flexor-extensor balance ratio of 2.2765.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S3, S4, S2

Evidence: High WAMP values observed in S3 (6325.3), S4 (2814.5), and S2 (2168.6) indicate significant motor unit recruitment consistent with high-effort sustained contraction.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.

sEMG Muscle Activation Report

Participant P9 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S2
Weakest Sensors	S8
Assessment Confidence	0.95

Clinical Summary: The session demonstrates a sustained, flexor-dominant contraction pattern. High co-contraction (0.55) suggests a power-grip style activation. S2 and S3 are the primary contributors to force production. S1 exhibits inconsistent activation compared to other sensors.

Key Findings:
- High co-contraction ratio
- Flexor-dominant activation pattern
- S1 intermittent profile

Session Overview

Session Duration	1203.4 s (20.1 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.015197 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.012799	0.410	101.2	89.3	1657	High	Intermittent
S2	0.031093	0.397	106.0	99.2	6204	Very High	Sustained
S3	0.020936	0.144	101.4	91.1	4251	Very High	Sustained
S4	0.015532	0.129	79.1	64.5	1251	Very High	Ramp-up
S5	0.012339	0.190	125.3	111.9	1753	High	Sustained
S6	0.007643	0.172	130.4	113.1	827	Moderate	Sustained
S7	0.015205	0.119	114.2	104.1	1983	Very High	Sustained
S8	0.006031	0.166	111.0	91.6	126	Moderate	Ramp-up

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S1: High variability in intermittent activation

S2: Primary driver of sustained contraction

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S2	Primary contributor
Weakest Sensor	S8	Lowest activity
Dominant-to-Weakest Ratio	5.16×	Activation disparity
Concentration Index	0.161	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.994	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.257	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.553	Agonist/antagonist balance
Flexor/Extensor Balance	1.98	>1 = flexor-dominant

Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows significantly higher activation levels compared to extensor group (S5-S8), with a flexor-extensor balance ratio of 1.98.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S2, S3

Evidence: High WAMP values observed in S2 (6204.4) and S3 (4250.6) indicate significant motor unit recruitment during sustained effort.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.